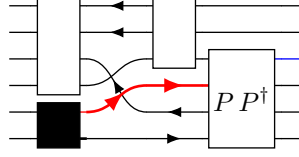


1 Birdtrack projectors

The **projector** environment is set up with the structure that symmetrisers and antisymmetrisers can appear under each other, separated by any number of **freelines**, wrapped in a **layer**. In-between layers carrying symmetrisers, antisymmetrisers and the more agnostic **operator** rectangles, permutations between lines can be added in a layer, with a **permutation**.

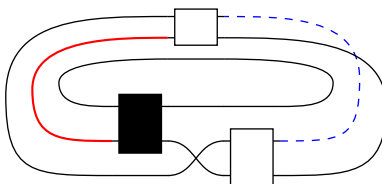


The projector in Eq. (1) is rendered from the source code:

```
\begin{projector}[permutation min width=1.75em]
  \startnodes                                     % short lines to the right/left
  \layer{
    \symmetriser{4}                                % number of lines
    \antisymmetriser
    []                                              % left lines formatting
    [{draw=red,line width=1pt}]                  % right lines formatting
    {2}                                            % number of lines
  }
  \layer{
    \permute
    [<,<,<,,{>},draw=red,line width=1pt],]      % < and > are arrow directions
    {1,2,3,4,5,6}                                % permutation domain
    {1,2,5,3,4,6}                                % permutation image
  }
  \layer{
    \symmetriser{3}
    \freelines                                    % straight lines below the S
    [{>,draw=red,line width=1pt},{<,>}]          % line formatting
    {3}                                            % number of free lines
  }
  \layer{
    \freelines{2}
    \operator                                        % a rectangle that extends
    [, {draw=red,line width=1pt}]                % itself to the text written
    [{blue}]                                      % inside.
    {4}                                            % rectangle height
    {\(\displaystyle \, P\, P^\dagger\, \)}      % text
  }
  \endnodes[,,{blue}]
\end{projector}
```

1.1 Traced projectors

We include the ability to draw a traced projector by breaking the object down into four components; a `leftconnect`, `topprojector`, `bottomprojector` and `rightconnect`. The top and bottom projectors are wrappers for regular projector objects. The top and bottom projectors can have different (anti)symmetriser widths, as well as different line spacings. Left- and right-connect layers will take care of merging the lines. On these connection layers, we can permute the lines to include line crossings. All lines can be typeset to have individually different colours, widths, arrows, etc.



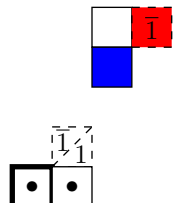
(2)

Equation 2 is typeset from the source code:

```
\begin{tracedprojector}[
  trace_gap=1em,                % gap between top and bottom projector
  trace_reach=4em,              % how far trace extends to the left/right
  trace_turn=4em,               % turn radius (higher is rounder)
  trace_lane_spacing=1.0em      % line spacing between lines on left/right-connect
]
\leftconnect{                  % trace lines on the left
  \permute
  [{draw=red,line width=.8pt},]
  {1,2,3}{3,2,1}
}
\topprojector[line_spacing=.8em]{ % top projector (small line spacing)
  \begin{projector}
    \layer{
      \symmetriser{2}\freelines{1}
    }
  \end{projector}
}
\bottomprojector[line_spacing=1.3em]{ % bottom projector (larger line spacing)
  \begin{projector} % typeset like a regular projector
    \layer{
      \antisymmetriser{2}\freelines{1}
    }
    \layer{
      \permute{1,2,3}{1,3,2}
    }
    \layer{
      \freelines{1}
      \symmetriser{2}
    }
  \end{projector}
}
\rightconnect{                  % trace lines to the right
  \permute
  [{draw=blue,dashed},]
  {1,2,3}{3,1,2}
}
\end{tracedprojector}
```

2 Young diagram pairs

Young diagrams and Young diagram pairs can be typeset in the `ydpair` environment. A `ydpair` can have either a `covar` (covariant) and/or a `convar` (contravariant) component. The distinction lies in the orientation; `covar` renders left- and top-aligned boxes, and `convar` renders bottom- and right-aligned boxes, following the pair-multiplication convention. Both parts can be padded by any integer number of boxes by the `hpad` command. Boxes can be typeset with arbitrary content, filling, line style and width.


(3)

Equation 3 has been typeset with the following code:

```
\begin{ydpair}
  \covar{
    [] & [fill=red,dashed] \ensuremath{\overline{1}} \\
    [fill=blue]
  }
  \hpad{1}
  \convar{
    [dashed] \splitbox{\overline{1}}{1} \\
    [line width=0.2em] \ensuremath{\bullet} & [] \ensuremath{\bullet}
  }
\end{ydpair}
```

3 Global configurations

In the preamble, `\birdtracksetup` admits parameters that set the document-wide style of birdtrack projectors and Young diagrams/pairs drawn with `birdtracks.sty`. The globally-configurable parameters are listed in Table 1.

Table 1: Configurable global `birdtracks.sty` style variables.

Setting	Default	Description
Projector settings		
layer padding	0.25em	Padding around composed layers
boundary length	0.1em	Length of exposed boundary lines
permutation min width	0pt	Minimum width of a permutation layer
stitch overlap	0.1pt	Overlap used to hide connection seams
arrow type	Latex	TikZ arrow-tip style
arrow size	1	Arrow-tip scale
Traced operator settings		
trace gap	2em	Gap between traced projectors
trace reach	1.5em	Outward reach of trace paths
trace turn	0.75em	Offset used at trace turns
trace lane spacing	0.55em	Spacing between nested trace paths
Custom global operator settings		
operator width	1.25em	Operator width
operator line separation	1.1em	Space between operator lines
operator line length	1.1em	Length of external operator lines
operator line width	0.5pt	Operator line thickness
operator border width	0.5pt	Operator border thickness
operator fill	white	Operator background colour
operator line colour	black	Operator line and border colour
operator text colour	black	Operator label colour
operator top margin	0.35	Top margin, in line separations
operator bottom margin	0.35	Bottom margin, in line separations
operator options	empty	Extra TikZ options for the operator
Young diagram pair settings		
box size	1.8em	Sets the <code>ydpair</code> box size
pair line width	0.5pt	<code>ydpair</code> border thickness
pair line colour	black	<code>ydpair</code> border colour
pair fill	white	<code>ydpair</code> background colour
pair text colour	black	<code>ydpair</code> label colour
pair text size	<code>\normalsize</code>	<code>ydpair</code> label size

4 Local configurations

Environment options apply only to the environment where they are set. The projector options can also be set on `topprojector` and `bottomprojector`. A `tracedprojector` accepts the global settings in Table 1 as local overrides.

Table 2: Configurable local `birdtracks.sty` settings.

Setting	Scope	Description
Projector environment settings		
<code>operator width</code>	<code>projector</code>	Minimum operator width
<code>line spacing</code>	<code>projector</code>	Space between diagram lines
<code>line width</code>	<code>projector</code>	Diagram-line thickness
<code>layer padding</code>	<code>projector</code>	Padding around layers
<code>permutation min width</code>	<code>projector</code>	Minimum permutation-layer width
<code>arrow type, arrow size</code>	<code>projector</code>	Arrow-tip style and scale
<code>stitch overlap</code>	<code>projector</code>	Hides seams between layers
<code>boundary length</code>	<code>projector</code>	Exposed boundary-line length
<code>operator border width</code>	<code>projector</code>	Operator border thickness
<code>operator top margin</code>	<code>projector</code>	Top margin, in line spacings
<code>operator bottom margin</code>	<code>projector</code>	Bottom margin, in line spacings
<code>operator fill, operator line colour, operator text colour</code>	<code>projector</code>	Operator fill, line, and label colours
Young diagram pair settings		
<code>box size</code>	<code>ydpair</code>	Size of each diagram box
<code>line width</code>	<code>ydpair</code>	Box-border thickness
<code>line colour</code>	<code>ydpair</code>	Box-border colour
<code>fill</code>	<code>ydpair</code>	Default box fill
<code>text colour</code>	<code>ydpair</code>	Default label colour
<code>text size</code>	<code>ydpair</code>	Default label size
Entry and command-level styles		
TikZ node options	<code>ydpair</code>	Style each <code>\covar</code> or <code>\convar</code> box independently, e.g., <code>[fill=blue!10]</code>
<code>\hpad{n}</code>		Fractional box-size gap between the diagrams
TikZ line options	Projector layer	Used by <code>\freelines</code> , <code>\permute</code> , <code>\startnodes</code> , and <code>\endnodes</code> ; a leading <code><</code> or <code>></code> adds an arrow
Up to two line-option lists	Projector layer	Used by <code>\symmetriser</code> , <code>\antisymmetriser</code> , and <code>\operator</code> for left- and right-side lines